

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Traversal Challenge

Course Code:	CSA0305	Course Title:	Data Structures Assessment Tool 2 – Traversal Challenge 100
CO Assessed:	CO3 – Precision (BL3, BL5)	Assessment:	
Weightage:	30% of CIA	Total Marks:	
CO3 Statement:	Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, BTrees, Red-Black Trees, and Splay Trees.		
Student Name:	SUMETHA SRI.S	Reg. No.:	192521097 03/08/2026
Section / Batch:		Date:	

Code of Conduct

I SUMETHA SRI.S(192521097) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ Sumetha _____

Instructions

- This assessment tool contains 5 Traversal Challenge questions, Total: 100 Marks.
- Students can use any online/offline Traversal Challenge. Demonstrate tree.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic	Focus	Q Nos	Marks
Tree Traversals	Traversal types, In order, Post order and Pre order	5	100
GRAND TOTAL:		100 Marks	

Traversal Challenge

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	20	Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Shows good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Traversal Accuracy	20	Correctly completes all tree traversals (Preorder, Inorder, Postorder, Level Order) without errors.	Completes most traversals correctly with one minor error.	Correctly performs only some traversals with multiple errors.	Unable to perform most traversals correctly.

Selection of Appropriate Traversal	30	Correctly selects the appropriate traversal for every given problem scenario.	Selects the correct traversal for most scenarios.	Selects appropriate traversal for only a few scenarios.	Frequently selects incorrect traversal methods.
Completion & Time Efficiency	20	Completes all challenges within the allotted time while maintaining accuracy.	Completes most challenges within the time limit.	Completes only part of the challenge or exceeds the time limit slightly.	Unable to complete the challenge within the allotted time.
Evidence Submission	10	Uploads complete screenshots showing successful completion, score, and student details.	Uploads screenshots with minor missing information.	Uploads incomplete or unclear screenshots.	No valid evidence submitted.

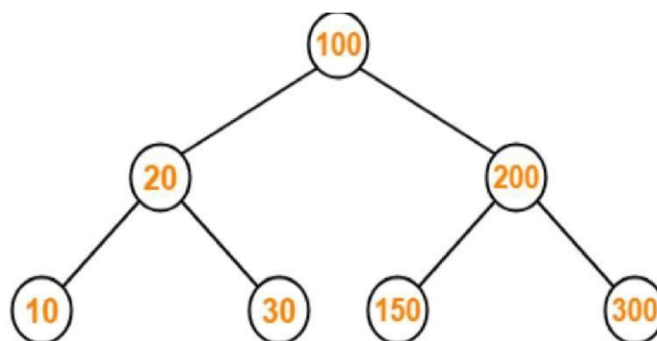
Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____

CSA0305 – Data Structures (Slot A)

CO3 - AT2 - Tree Traversal Challenges **Question**

No:1

Write the traversal sequences for following trees

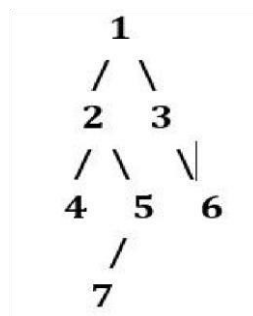


Question No: 2

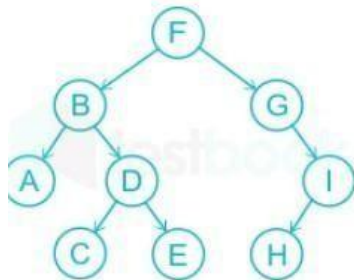
Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?

Question No: 3

Write the Inorder, Preorder & Postorder traversal sequences for following trees

**Question No: 4**

What is the sequence of nodes when applying in-order traversal on the binary tree given below?



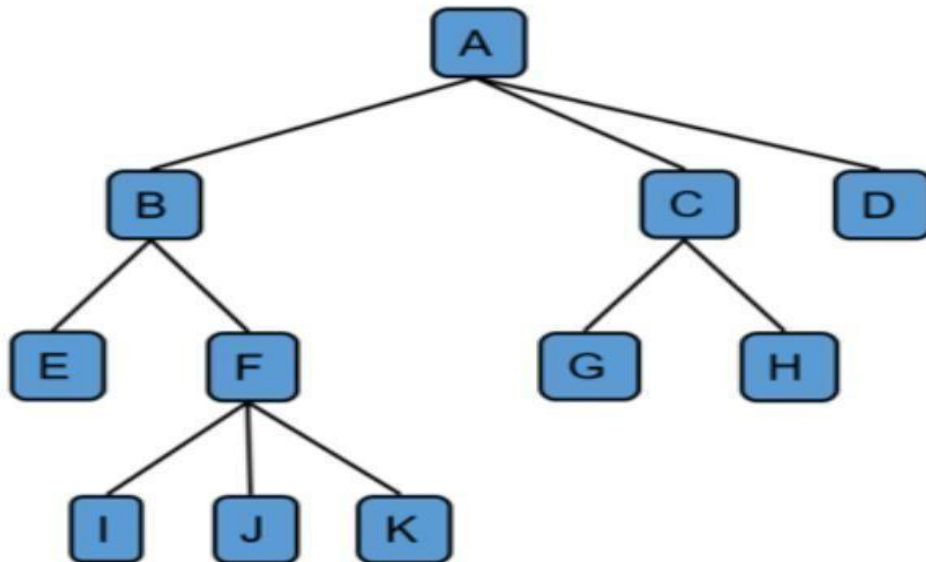
1. A, B, C, D, E, F, I, H, G

2. A, C, D, E, B, F, G, H, I

3. A, B, C, D, E, F, G, H, I

4. I, H, G, F, E, D, C, B, A

Question No: 5



QUESTION 1

PREORDER

[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

100,20,200,10,30,150,300

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

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[← Back to Home](#)

Tree Traversals



Input

Input is level-by-level order.

100,20,200,10,30,150,300

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

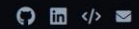
Visualize

Start Simulation

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[← Back to Home](#)

Tree Traversals



Input

Input is level-by-level order.

100,20,200,10,30,150,300

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

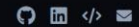
Visualize

Start Simulation

100 20 10 30 200 150 300



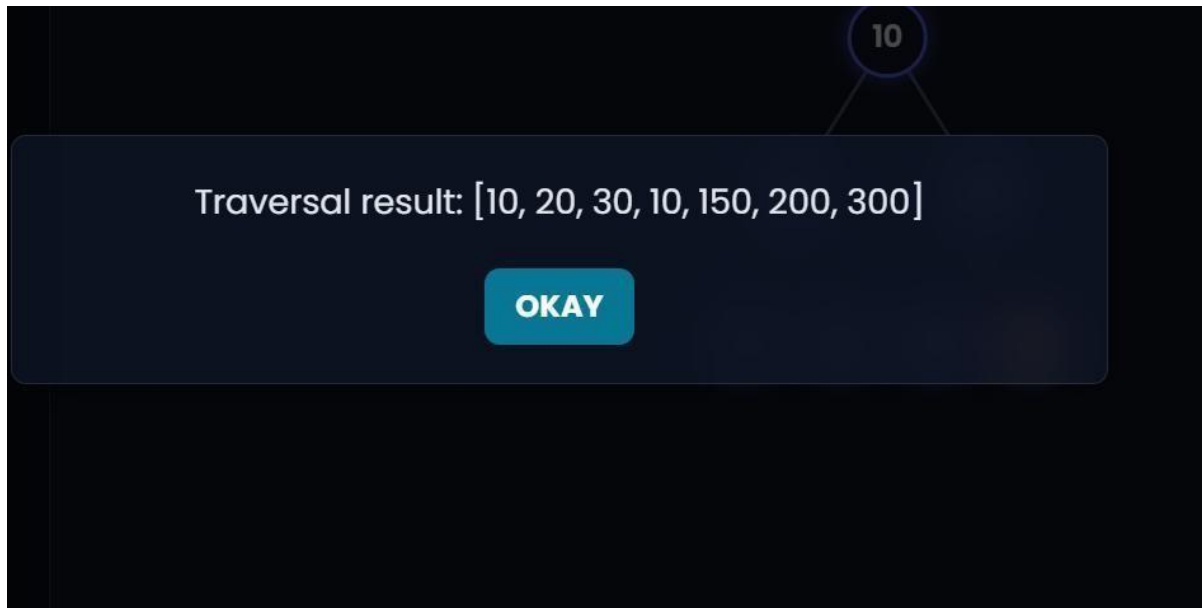
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Traversal result: [10, 20, 10, 30, 200, 150, 300]

OKAY

INORDER



POST ORDER

[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

10,20,200,10,30,150,300

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

10

20

200

10

30

150

300

[← Back to Home](#)

Tree Traversals

?

Input

Input is level-by-level order.

100,20,200,10,30,150,300

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

103020150300200100

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Traversal result: [10, 30, 20, 150, 300, 200, 10]

OKAY

QUESTION 2

PREORDER

Input
Input is level-by-level order.

7, 5, 8, 1, 6, null, 9, 0, 3, null, null, null, null, 2, 4

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

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[← Back to Home](#)

Tree Traversals

?

Input
Input is level-by-level order.

7, 5, 8, 1, 6, null, 9, 0, 3, null, null, null, null, 2, 4

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

7

5

8

1

6

9

0

3

2

4

[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

7, 5, 8, 1, 6, null, 9, 0, 3, null, null, null, null, 2, 4

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

2 4 0 3 1 6 5 9 8 7

7

5

8

1

6

0

3

2

4

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5

Traversal result: [7, 5, 1, 0, 2, 4, 3, 6, 8, 9]

OKAY

2

4

INORDER

Tree Traversals

Input

7, 5, 8, 1, 6, null, 9, 0, 3, null, null,
null, null, 2, 4

Traversal Order

Pre

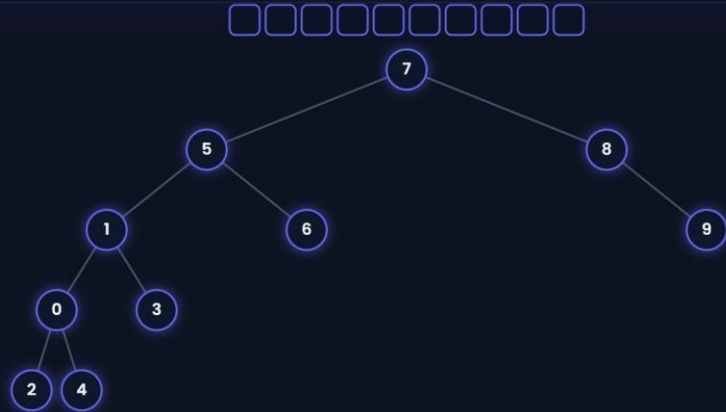
In

Post

 Generate Default

► **Visualize**

▶ Start Simulation



Tree Traversals

Input

Input is level-by-level order.

7, 5, 8, 1, 6, null, 9, 0, 3, null, null,
null, null, 2, 4

Traversal Order

Pre

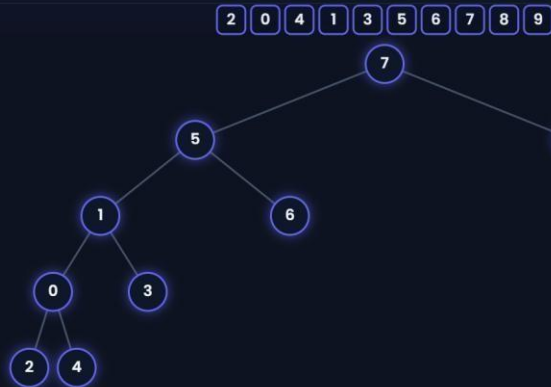
In

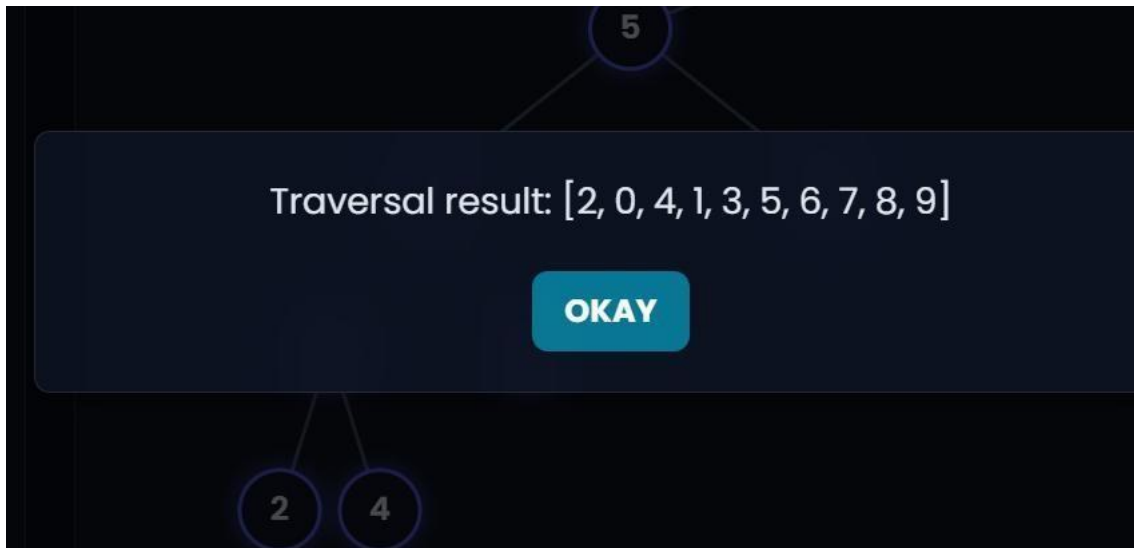
Post

 Generate Default

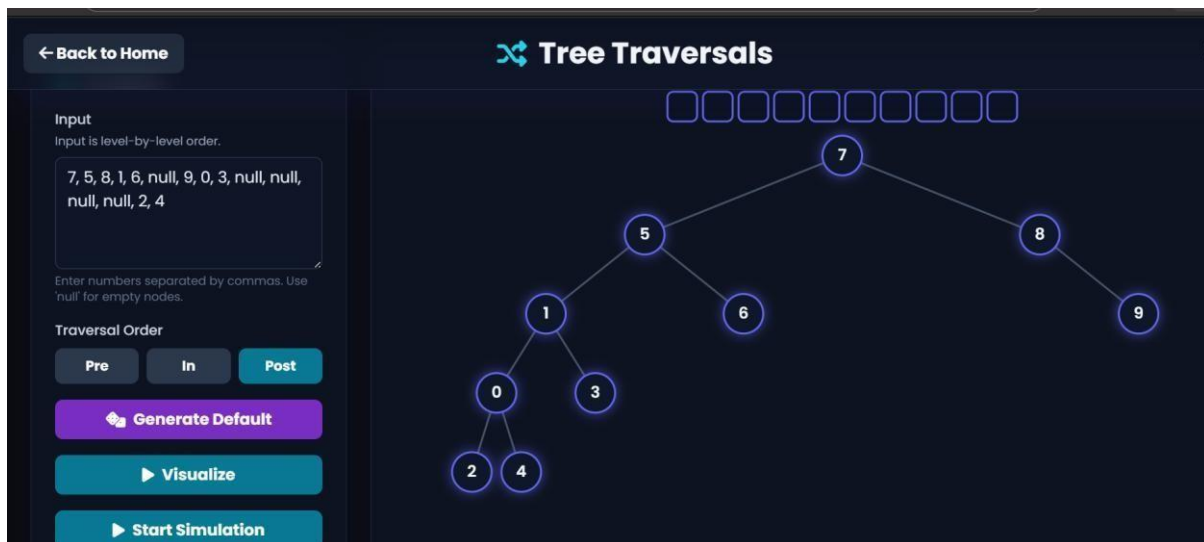
► **Visualize**

▶ Start Simulation





POSTORDER



[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

7, 5, 8, 1, 6, null, 9, 0, 3, null, null, null, null, 2, 4

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

2 4 0 3 1 6 5 9 8 7

7

5

8

1

6

0

3

2

4

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5

Traversal result: [2, 4, 0, 3, 1, 6, 5, 9, 8, 7]

OKAY

2

4

QUESTION 3

PREORDER

[← Back to Home](#)

Tree Traversals

?

Input
Input is level-by-level order.

1, 2, 3, 4, 5, null, 6, null, null, 7,
null, null, null, null, null

Enter numbers separated by commas. Use
null for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

Start Simulation

```
graph TD; 1((1)) --- 2((2)); 1 --- 3((3)); 2 --- 4((4)); 2 --- 5((5)); 5 --- 7((7)); 3 --- 6((6));
```

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[← Back to Home](#)

Tree Traversals

?

Input
Input is level-by-level order.

1, 2, 3, 4, 5, null, 6, null, null, 7,
null, null, null, null, null

Enter numbers separated by commas. Use
null for empty nodes.

Traversal Order

Pre In Post

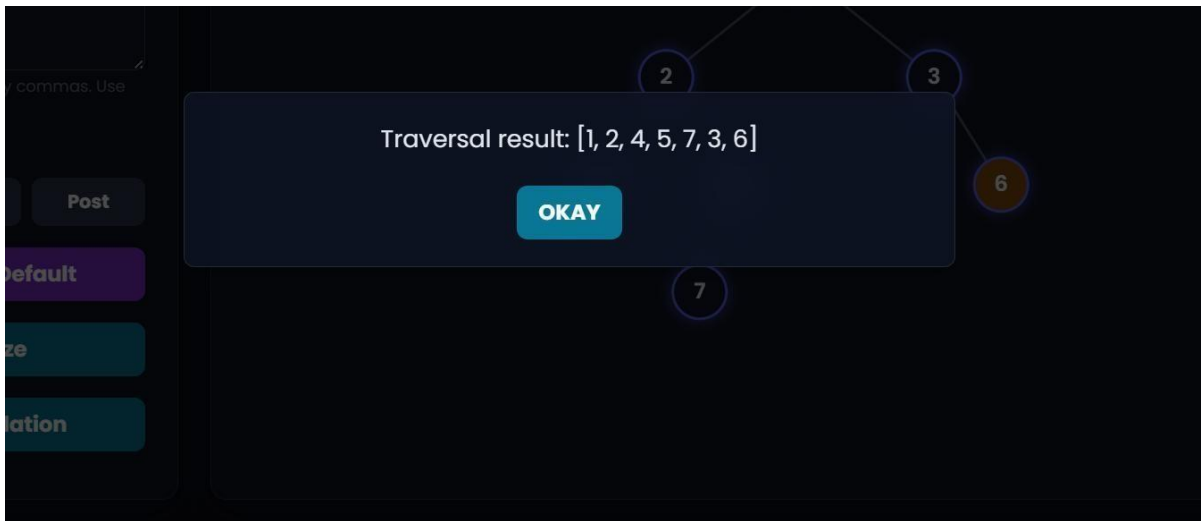
Generate Default

Visualize

Start Simulation

```
graph TD; 1((1)) --- 2((2)); 1 --- 3((3)); 2 --- 4((4)); 2 --- 5((5)); 5 --- 7((7)); 3 --- 6((6));
```

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INORDER

[← Back to Home](#)

Tree Traversals

Input
Input is level-by-level order.

1, 2, 3, 4, 5, null, 6, null, null, 7, null, null, null, null, null

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

```
graph TD; 1((1)) --- 2((2)); 1 --- 3((3)); 2 --- 4((4)); 2 --- 5((5)); 5 --- 7((7)); 3 --- 6((6));
```

[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

1, 2, 3, 4, 5, null, 6, null, null, 7, null, null, null, null, null

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

4 2 7 5 1 3 6

1

2

3

4

5

6

7

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1

2

3

4

5

6

7

Traversal result: [4, 2, 7, 5, 1, 3, 6]

OKAY

POSTORDER

Tree Traversals

Input

Input is level-by-level order.

1, 2, 3, 4, 5, null, 6, null, null, 7,
null, null, null, null, null

Traversal Order

Pre

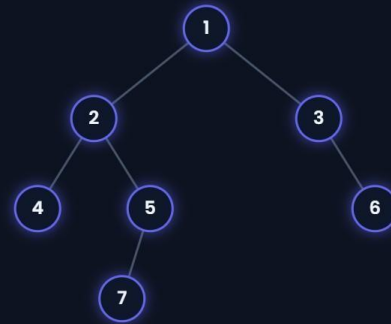
In

Post

 Generate Default

► **Visualize**

▶ Start Simulation



Tree Traversals

Input

Input is level-by-level order.

1, 2, 3, 4, 5, null, 6, null, null, 7,
null, null, null, null, null

Traversal Order

Pre

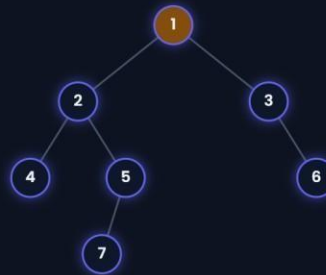
In

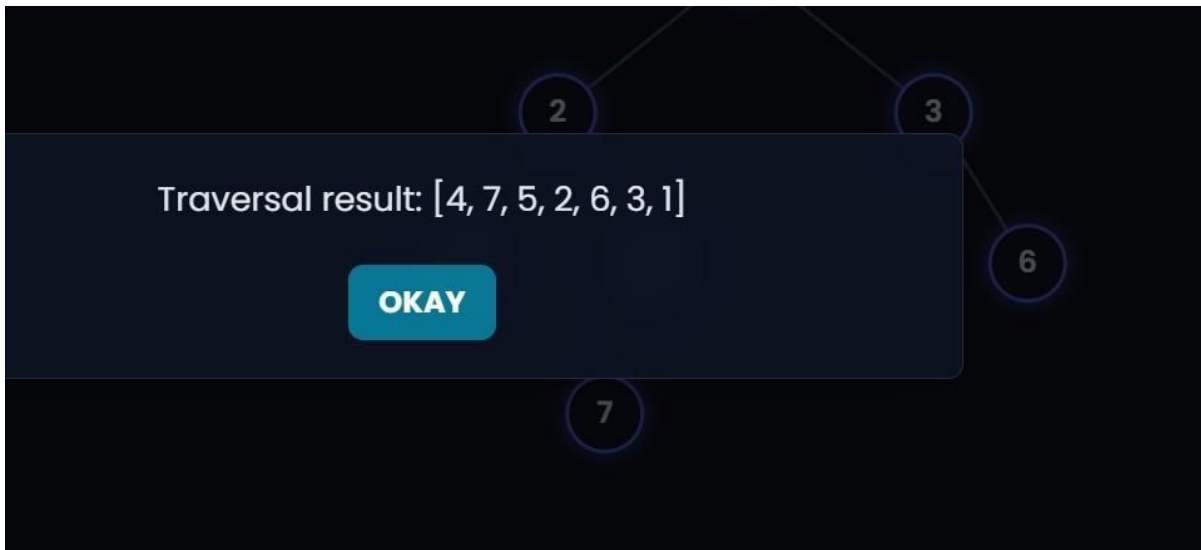
Post

 Generate Default

► **Visualize**

▶ Start Simulation





QUESTION 4

INORDER

[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

6, 2, 7, 1, 4, null, 9, 3, 5, null, null, null, 8, null, null

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

6

2

7

1

4

9

8

3

5

[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

6, 2, 7, 1, 4, null, 9, 3, 5, null, null, null, 8, null, null

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

315246798

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27

Traversal result: [3, 1, 5, 2, 4, 6, 7, 9, 8]

OKAY

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3,1,5,2,4,6,7,9,8

A,B,C,D,E,F,G,H,I

QUESTION 5

PREORDER

POSTORDER

[← Back to Home](#)

Tree Traversals

Input
Input is level-by-level order.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

```
graph TD; 1((1)) --- 2((2)); 1 --- 3((3)); 2 --- 4((4)); 2 --- 5((5)); 3 --- 6((6)); 3 --- 7((7)); 4 --- 8((8)); 4 --- 9((9)); 5 --- 10((10)); 5 --- 11((11));
```

[← Back to Home](#)

Tree Traversals

8 9 4 10 11 5 2 6 7 3

Input
Input is level-by-level order.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

```
graph TD; 1((1)) --- 2((2)); 1 --- 3((3)); 2 --- 4((4)); 2 --- 5((5)); 3 --- 6((6)); 3 --- 7((7)); 4 --- 8((8)); 4 --- 9((9)); 5 --- 10((10)); 5 --- 11((11));
```

